Internal Assessment 1 - Engineering Physics

Semester I | Time: 1 hour | Maximum marks: 20

Answer ALL questions. Draw neat, labelled diagrams wherever necessary.

Q1. Newton's laws of motion **[5 marks]**

State Newton's three laws of motion. Give one everyday example of the third law.

Q2. Uniformly accelerated motion **[5 marks]**

A car starts from rest and accelerates uniformly to 20 m/s in 5 s.

(a) Calculate its acceleration. [2]

(b) Calculate the distance it travels in this time. [3]

Q3. Ohm's law **[5 marks]**

State Ohm's law. Draw the V-I graph for an ohmic conductor and explain how the resistance can be found from the graph.

Q4. Energy **[5 marks]**

Define kinetic energy and potential energy. A ball of mass 2 kg is dropped from rest from a height of 20 m. Find its speed just before it hits the ground. Take g = 10 m/s^2 and neglect air resistance.